

A 本文所用全部测试集内容

为了较为全面地考察大语言模型在化学核心任务上的推理表现，本研究自行设计并整理出了一套规模在 700 道题以上的综合化学测试集。该测试集在覆盖面上较为宽泛，既涉及从简单无机物到结构复杂的分子等不同类型的化学物质，也兼顾了从一维字符串到三维晶体结构等多种表达层级。为便于后续工作的复现与参考，下文将本文九项核心任务所对应的全部测试内容统一汇总于此：

A.1 分子名称与 SMILES 代码的双向转换测试集

Methane; Ethane; Propane; Butane; Ethene; Propene; Acetylene; Propyne; Methanol; Ethanol; Formaldehyde; Acetaldehyde; Formic acid; Acetic acid; Propionic acid; Acetone; Diethyl ether; Methylamine; Ethylamine; Acetonitrile; Dimethylamine; Isobutane; Neopentane; Isopropanol; Tert-Butanol; Triethylamine; Chloromethane; Dichloromethane; Chloroform; Carbon tetrachloride; Methyl tert-butyl ether; Benzene; Naphthalene; Anthracene; Phenanthrene; Toluene; Ethylbenzene; Phenol; Aniline; Nitrobenzene; Benzoic acid; Benzaldehyde; Cyclopropane; Cyclobutane; Cyclopentane; Cyclohexane; Cyclohexene; Pyridine; Pyridazine; Pyrimidine; Pyrazine; Pyrrole; Imidazole; Pyrazole; Furan; Thiophene; Oxazole; Isoxazole; Thiazole; Piperidine; Morpholine; Ethylene glycol; Glycerol; Urea; (*Z*)-But-2-ene; (*E*)-But-2-ene; (*Z*)-1,2-Dichloroethene; (*E*)-1,2-Dichloroethene; Maleic acid; Fumaric acid; (*R*)-Butan-2-ol; (*S*)-Butan-2-ol; (*R*)-Limonene; (*S*)-Limonene; (*R*)-Ibuprofen; (*S*)-Ibuprofen; (*R*)-Thalidomide; (*S*)-Thalidomide; (+)-Tartaric acid; (-)-Tartaric acid; *meso*-Tartaric acid; D-alanine; L-alanine; L-Leucine; D-Leucine; L-Cysteine; D-Cysteine; L-Fructose; D-Fructose; L-Malic acid; D-Malic acid; L-Glyceraldehyde; D-Glyceraldehyde; Cholesterol; Testosterone; Estradiol; Progesterone; Dexamethasone; Cortisone; Caffeine; Nicotine; Morphine; Codeine; Heroin; Quinine; Aspirin; Penicillin G; Doxorubicin; Vincristine; Paclitaxel; Erythromycin A; Streptomycin; Tetracycline; Tetrodotoxin; Saxitoxin; Sorafenib; Imatinib; Adamantane; Norbornane; Bicyclo[2.2.2]octane; Camphor; Borneol; alpha-Pinene; Spiro[4.5]decane; Spiro[2.2]pentane; Spiro[3.3]heptane; 1,3-Cyclohexanedione; Fullerene C60; 18-Crown-6; [2.2.2]Cryptand; Sodium acetate; Ammonium chloride; Copper(II) sulfate; Silver nitrate; Ethylmagnesium bromide; Tetrabutylammonium hydroxide; Deuterium oxide; Deuterated acetone; [¹³C]Methane; Tritiated benzene; [¹⁵N]Glycine; [¹⁴C]Acetic acid (carboxy-labeled); [¹⁴C]Acetic acid (methyl-labeled); Phenanthrene-D10; p-Terphenyl-d14 以及上述分子对应的 SMILES 字符串。

A.2 辛醇-水分配系数 ($\log P$) 预测测试集

Methane; Ethane; Propane; Butane; Pentane; Hexane; Heptane; Octane; Methanol; Ethanol; 1-Propanol; 1-Butanol; 1-Pentanol; 1-Hexanol; Formic acid; Acetic acid; Propanoic acid; Butyric acid; Benzene; Toluene; Ethylbenzene; Naphthalene; Biphenyl; Anthracene; Acetone; Butanone; Acetaldehyde; Acetophenone; Diethyl ether; Tetrahydrofuran; Anisole; Methylamine; Triethylamine; Aniline; Chloroform; Carbon tetrachloride; Chlorobenzene; Bromobenzene; Iodobenzene; Pyridine; Ethyl acetate; Phenol; Nitrobenzene; Cyclohexane; Glycine; Alanine; Valine; Phenylalanine; Tryptophan; Aspartic Acid; Lysine; Ibuprofen; Aspirin; Sertraline; Fluoxetine; Atenolol; Metformin; Indigo;

beta-Carotene; linalool; limonene; gamma-Terpinene; (+)-Camphene; camphor; eugenol; citronellal; citronellol; p-cymene; (*R*)-(-)-Carvone; (1*R*)-(-)-fenchone; geraniol; carvacrol; thymol; alpha-terpinene; (1*R*)-Camphor; Eucalyptol; L-(-)-menthol; (-)-menthone; (-)-*cis*-Myrtanylamine; (1*S*)-(-)-Verbenone; Linalool oxide; Piperitone; 1,4-Cineole; Myrtenal; (-)-Borneol; Dihydrocarvone; (+)-Pulegone; (-)-Carveol; (-)-Perilla aldehyde; (*S*)-*cis*-Verbenol; (+)-Fenchol; (+)-Isomenthol; Limonene oxide; Dihydrocarveol; Terpinen-4-ol; alpha-Terpineol; Methyleugenol; Nerol; beta-Ionone; Linalyl acetate; Neryl acetate; Menthyl acetate; Geranyl acetate; alpha-Terpinolene; Car-3-ene; (+)-alpha-Pinene

A.3 极性表面积 (PSA) 预测测试集

diazepam; oxazepam; temazepam; lorazepam; nitrazepam; clonazepam; flurazepam; flutoprazepam; estazolam; midazolam; zopiclone; zolpidem hemitartrate; ramelteon; barbituric acid; Amobarbital; pentobarbital; phenobarbital; secobarbital; hexobarbital; phenytoin; ethotoin; fosphenytoin; trimethadione; dimethadione; phensuximide; methsuximide; ethosuximide; carbamazepine; oxcarbazepine; Vigabatrin; gabapentin; tiagabine; sodium valproate; zonisamide; topiramate; lamotrigine; chlorpromazine; acetopromazine; perphenazine; fluphenazine; fluphenazine enanthate; phenazine decanoate; thioridazine; thiethylperazine; pipothiazine; trifluoperazine; chlorprothixene; zuclopenthixol; flupenthixol; thiotixene; haloperidol; trifluoperidol; spiperone; droperidol; timiperone; penfluridol; fluspirilene; pimozide; metoclopramide; sulpiride; remoxipride; nemonapride; tiapride; clozapine; clothiapine; olanzapine; quetiapine; risperidone; aripiprazole; loxapine; amoxapine; mosapramine; ziprasidone; moclobemide; reboxetine; amoxapine; maprotiline; desipramine; clomipramine; trimipramine; doxepin; amitriptyline; nortriptyline; fluoxetine; paroxetine; citalopram; sertraline; mirtazapine; venlafaxine; desvenlafaxine; duloxetine; milnacipran; bupropion; buspirone; chlormezanone; tandospirone; lithium carbonate; bromocriptine; apomorphine; pramipexole; ropinirole; talipexole; levodopa; carbidopa; benserazide; selegiline; rasagiline; safinamide; entacapone; tolcapone; nebicapone; memantine; amantadine; riluzole; tacrine; donepezil; rivastigmine tartrate; galanthamine; morphine; codeine; heterocodeine; thebaine; etorphine; naloxone; naltrexone; levorphanol; butorphanol; pentazocine; pethidine; anileridine; fentanyl; alfentanil; sufentanil; carfentanil; remifentanil; methadone; dextromoramide; dextropropoxyphene

A.4 分子几何构型判定测试集

BeCl₂; CH₄; PCl₅; SF₆; CO₂; BF₃; SiH₄; AsF₅; XeF₆; CCl₄; SbF₅; NH₃; H₂O; SO₂; SF₄; ClF₃; XeF₂; PH₃; H₂S; BrF₅; XeF₄; OF₂; PCl₃; BrF₃; XeO₃; ICl₃; NO₃⁻; SO₄²⁻; PO₄³⁻; ClO₃⁻; CO₃²⁻; NH₄⁺; I₃⁻; ICl₄⁻; O₃; NO₂⁻; N₃⁻; SO₃; SCN⁻; AsH₃; H₂Se; KrF₂; TeCl₄; TlCl₃

A.5 分子点群对称性信息识别测试集

2,3-Pentadiene; Triphenylphosphine oxide; H₂O; SCl₄; BrF₃; Cyclopentadiene; 4-chloropyridine; 1,5-Naphthyridine; 1,4-Cyclohexanedione; 1,3,5-Trioxane; Isobutane; NH₃; XeOF₅⁻; Acetylene; Hydrogen Cyanide; Quinoline; N₂O₃; *meso*-Tartaric Acid; 4,4'-Dichlorobiphenyl; Twistane; 1,4-benzoquinone; Naphthalene; Ethene; AsF₅;

PF₅; Fe(CO)₅; Cyclobutane; Tetraphenylene; Chair conformation of cyclohexane; S8; Staggered Ferrocene; Eclipsed Ferrocene; Benzene; Uranocene; Carbon dioxide; Oxygen; [Co(NO₂)₆]³⁻; P₄; Carbon tetrachloride; Adamantane; CH₄; SF₆; 1,3,5,7-Cyclooctatetraene; C₆₀

A.6 ¹H NMR 谱图信号数量预测测试集

Methane; Ethane; Propane; n-Butane; Isobutane; Neopentane; Cyclopropane; Cyclobutane; Cyclopentane; Cyclohexane; Chloromethane; Bromoethane; 1-Propanol; Isopropanol; Diethyl ether; Ethyl methyl ether; Acetone; Acetaldehyde; Acetic acid; Methyl acetate; Ethylamine; 1-Butene; 1-Butyne; Benzene; Toluene; tert-Butanol; Isopropylamine; Vinyl chloride; Propene; Lactic acid; Acrylic acid; 3-Hydroxypropionaldehyde; Ethyl acetoacetate; 1,2-Dichloroethane; 1,1-Dichloroethane; Ethylene glycol; p-Xylene; o-Xylene; m-Xylene; Acetophenone; Benzoic acid; Benzaldehyde; Cinnamic acid; 4-Nitrotoluene; 2-Nitrotoluene; 3-Nitrotoluene; Aniline; N,N-Dimethylformamide; Furfural; 1,1,2-Trichloroethane; *cis*-1,2-Dimethylcyclopropane; *trans*-1,2-Dimethylcyclopropane; Limonene; Camphor; Citral; 1,3-Dioxolane; Tetrahydrofuran; Piperidine; 2-Methyl-2-butene; 3,3-Dimethyl-1-butene; Styrene; Cyclohexene; 1,3-Butadiene; 1,4-Cyclohexadiene; Butanone; Diethyl ketone; Isobutyric acid; Acrylonitrile; Acetylsalicylic acid; (-)-pinene; Lycopene; β -Carotene; Cholesterol; Estrone; Quinine; Strychnine; Ascorbic acid; Sucrose; [2.2]Paracyclophane; Adamantane

A.7 材料 XRD 特征峰位置的预测测试集

Cu; Au; Ni; NaCl; MgO; TiO₂; CaCO₃; ZrO₂; CuAu; GdFeO₃; ZIF-8; UiO-66; MOF-5

A.8 晶体学信息文件 (CIF) 生成测试集

Cu; K; NaCl; GaN; CaCO₃; Benzene; Urea; Benzoic Acid; Aspirin; Ferrocene